

Mixture viscosity measurement --- Solution

W26 (2026) — English translation

- A1 Express the molar concentration of water n_w in solution in terms of the mass concentration of glycerol w_g , μ_w and μ_g . 0.50**

The amount of a substance in a mixture is defined as:

$$\nu = \frac{m}{\mu},$$

where m is the mass of the substance, and μ is its molar mass. Let's write the formula for the molar concentration of water:

$$n_w = 1 - n_g = 1 - \frac{\nu_g}{\nu_g + \nu_w},$$

from here:

Answer:

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)}$$

- A2 Using the equation (2), convert the values of T_g to the values of w_g . 2.00**

Using the table method or the method of simple iterations with $a = 0.9$, we get:

Answer: $T_g, K, w_g, \eta, \Pi a \cdot c$ 195.2 1.000 0.906 194.8 0.996 0.839 194.6 0.994 0.767 193.9 0.987 0.655 193.1 0.979 0.558 191.8 0.965 0.479 190.8 0.954 0.373 188.6 0.930 0.202 183.4 0.869 0.0674 173.0 0.728 0.0278 170.0 0.682 0.0134 154.6 0.384 0.0026 149.3 0.252 0.0017 143.8 0.100 0.0012 140.3 0 0.0009

- A3 Propose a linearization that describes a system with a low molar concentration of water, which can be used to determine the parameters ξ and k . 0.30**

Answer: The dependence $\ln \frac{\eta}{1 \Pi a \cdot c}(n_w)$ is linear.

- A4 Plot the linearized relationship over the range $n_w \in [0, 0.2]$. Determine the parameters ξ and k using the resulting graph. 1.20**

Let's recalculate the points into the dependence $\ln \frac{\eta}{1 \Pi a \cdot c}(n_w)$.

$T_g, K, w_g, \eta, \Pi a \cdot c, n_w, \ln \eta$ 195.2 1.000 0.906 0 -0.099 194.8 0.996 0.839 0.020 -0.176 194.6 0.994 0.767 0.030 -0.265 193.9 0.987 0.655 0.063 -0.423 193.1 0.979 0.558 0.099 -0.583 191.8 0.965 0.479 0.156 -0.736 190.8 0.954 0.373 0.198 -0.986 188.6 0.930 0.202 0.278 -1.600 183.4 0.869 0.0674 173.0 0.728 0.0278 170.0 0.682 0.0134 154.6 0.384 0.0026 149.3 0.252 0.0017 143.8 0.100 0.0012 140.3 0 0.0009

$$\ln \frac{\eta}{1 \Pi a \cdot c}(n_w)$$

Approximating the dependence of the straight line, we obtain:

$$k = 4.31,$$

$$\ln \frac{\xi}{1 \Pi a \cdot c} = -0.12,$$

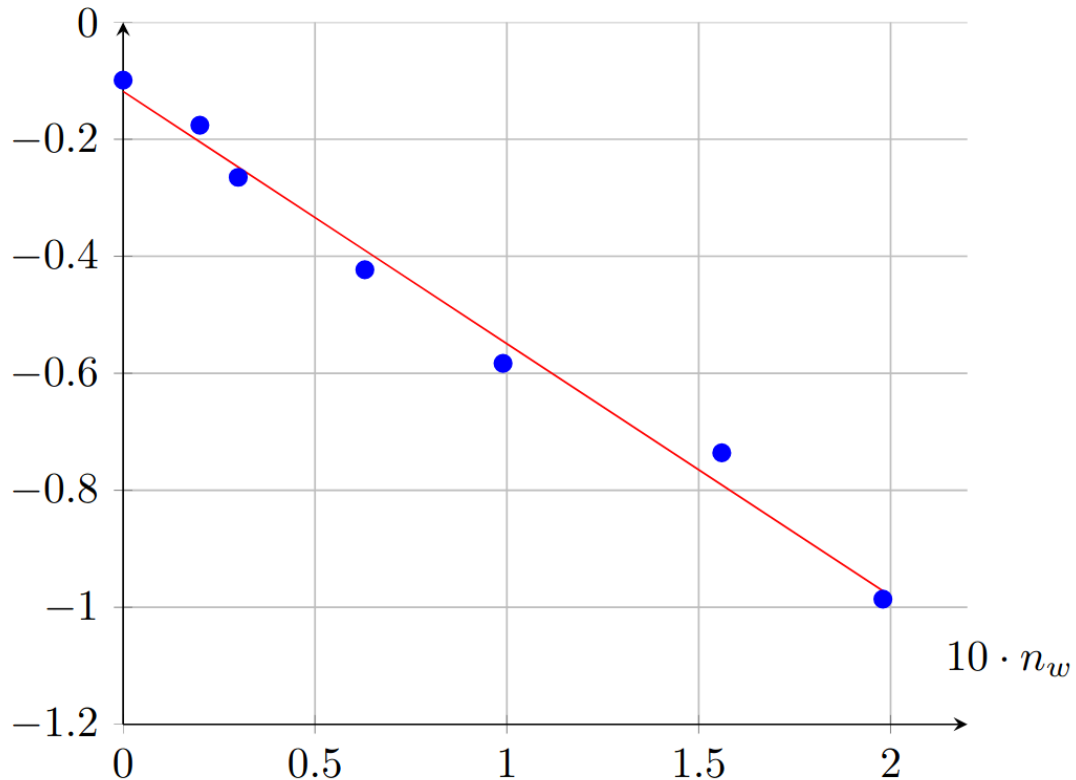
from which it follows:

Answer:

$$k = 4.31$$

$$\xi = 0.887 \Pi a \cdot c$$

$$\ln\left(\frac{\eta}{1 \Pi a \cdot c}\right)$$



A5 Knowing the parameters ξ and k , propose a linearization that allows you to determine the values of the parameters a and b . 0.30

Answer: The dependence $\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right| (n_w)$ is linear.

A6 Plot the linearized relationship. From the graph, determine a and b . Note: due to the inaccuracy of experimental data, some values cannot be recalculated. There are no more than 5 such values. 1.20

Note: due to the inaccuracy of experimental data, some values cannot be recalculated. There are no more than 5 such values.

Let's continue recalculating points:

T_g	K	w_g	η	$\Pi a \cdot c$	n_w	$\ln \left \frac{\xi \exp(-kn_w)}{\eta} - 1 \right $	195.2	1.000	0.906	0	-3.92	194.8	0.996	0.839	0.020	-3.54	194.6	0.994	0.767
0.030	-4.12	193.9	0.987	0.655	0.063	-3.44	193.1	0.979	0.558	0.099	-3.28	191.8	0.965	0.479	0.156	-2.91	190.8	0.954	0.373
0.198	-4.35	188.6	0.930	0.202	0.278	-1.12	183.4	0.869	0.0674	0.435	0.02	173.0	0.728	0.0278	0.656	-0.12	170.0	0.682	

0.0134 0.704 0.78 154.6 0.384 0.0026 0.891 1.85 149.3 0.252 0.0017 0.938 2.10 143.8 0.100 0.0012 0.979 2.29 140.3 0.0009 1 2.50

Let's plot the resulting points on the graph:

$$\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right| (n_w)$$

In the range $n_w \in [0, 0.2]$, the expression under the logarithm has a different sign and is near zero, so the function in this region behaves non-monotonic. To draw a straight line, we take into account only points in the range $n_w \in [0.7, 1]$.

$$\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right| (n_w) \text{ in the range } n_w \in [0.7, 1] \text{ (not required from participants)}$$

For $n_w > 0.7$, $\frac{\xi \exp(-kn_w)}{\eta} > 1$ holds, so $a > 0$. Having drawn the approximating straight line, we find:

$$b = 5.66,$$

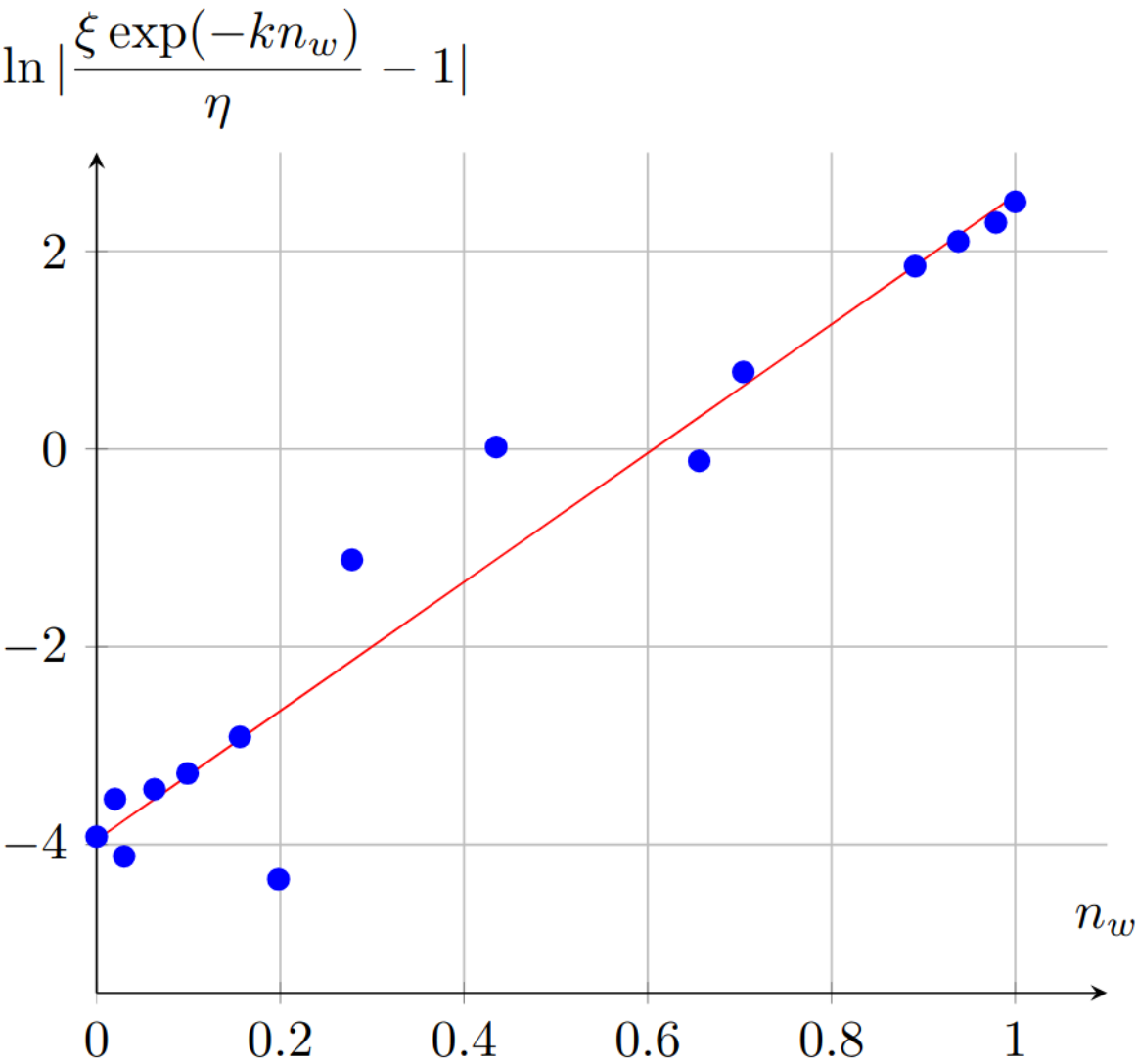
$$\ln a = -3.21,$$

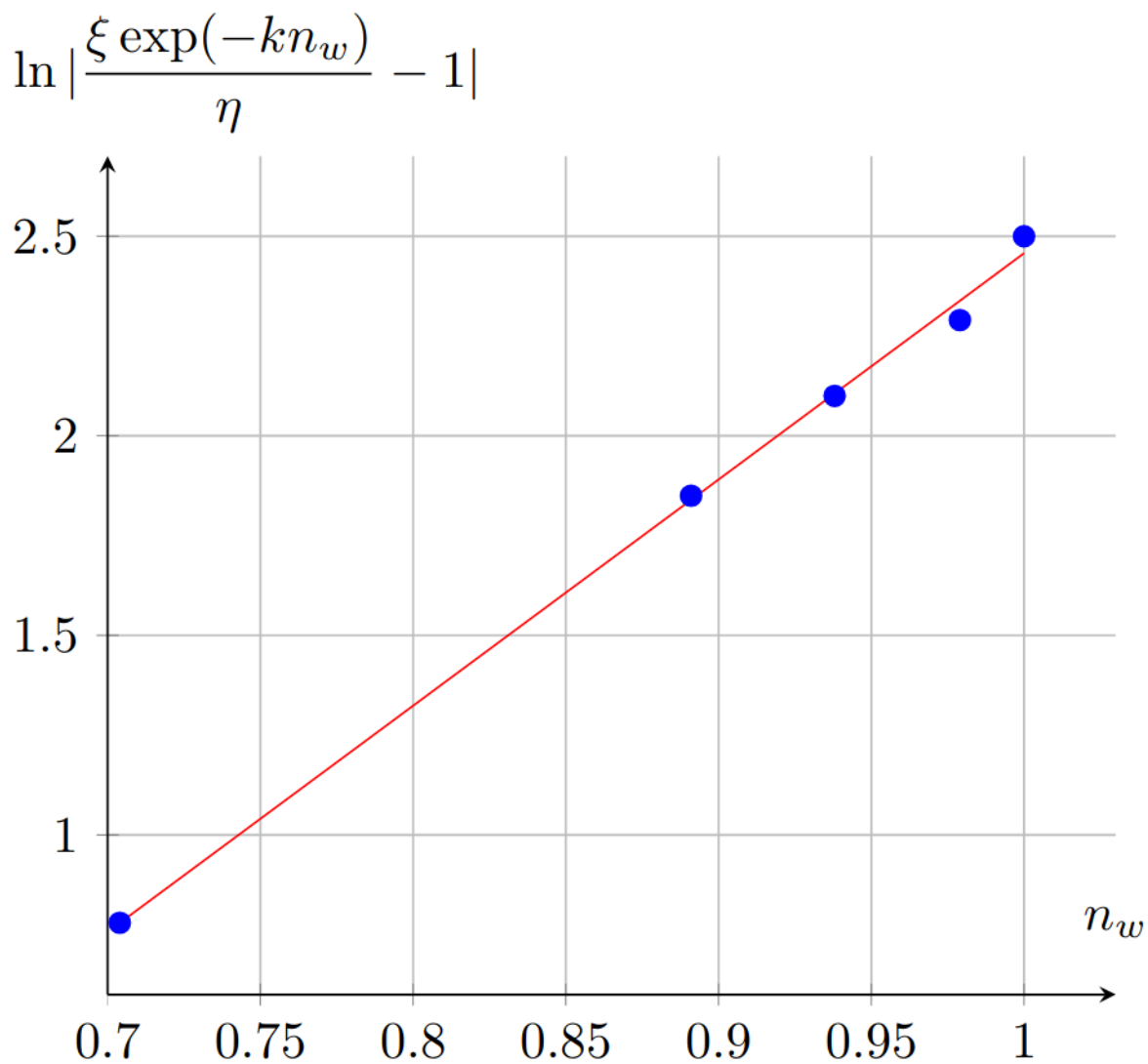
from here:

Answer:

$$a = 0.040$$

$$b = 5.66$$





A7 Draw a graph of the theoretical and experimental dependences $\eta(n_w)$ on one piece of graph paper.

1.00

Let's calculate the theoretical values of $\eta(n_w)$:

$T_g, K \quad w_g \quad \eta, \text{Па} \cdot \text{с} \quad n_w \quad \ln \eta \quad \ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right| \quad \eta_{\text{теор}}, \text{Па} \cdot \text{с}$

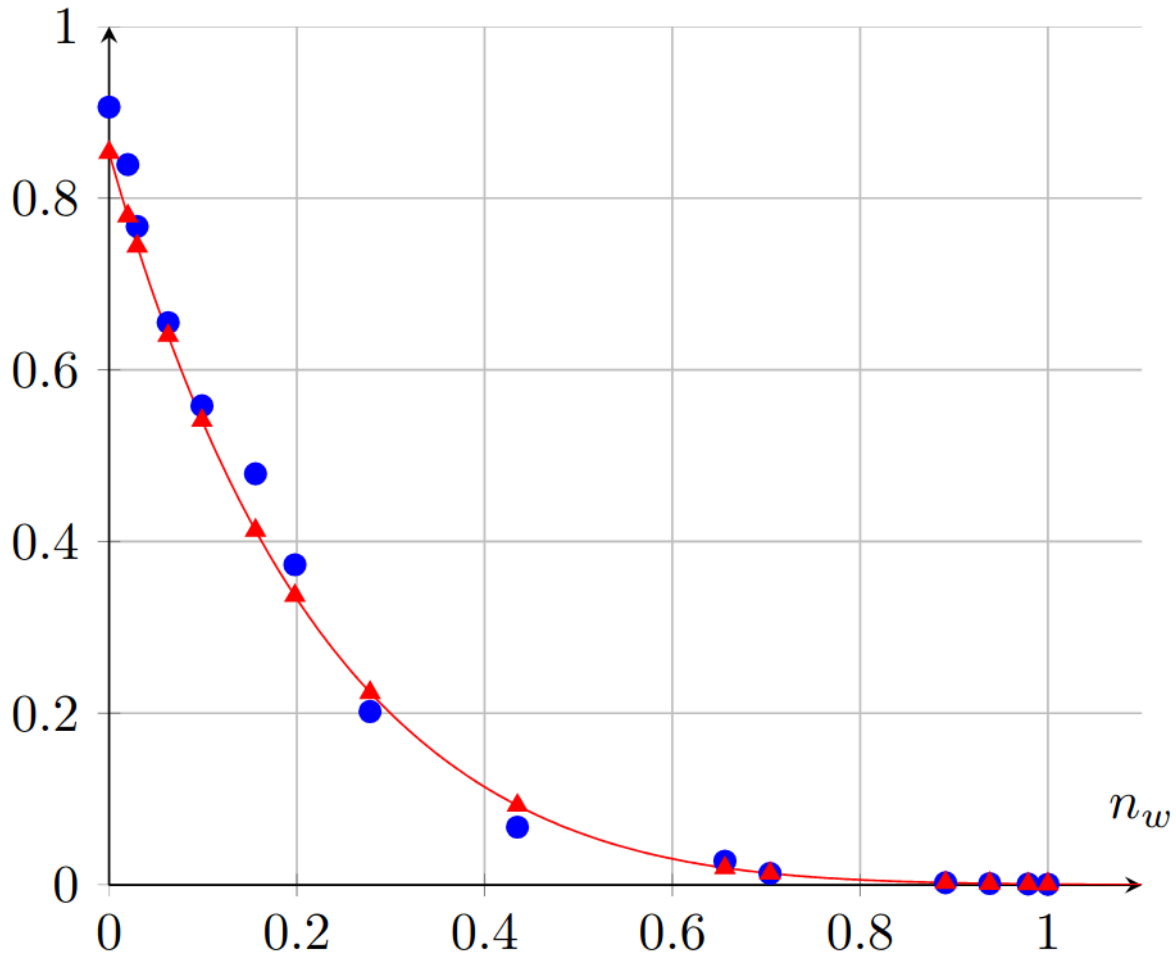
195.2	1.000	0.906	0	-0.099	-3.92	0.853	194.8	0.996	0.839
0.020	-0.176	-3.54	0.779	194.6	0.994	0.767	0.030	-0.265	-4.12
0.744	193.9	0.987	0.655	0.063	-0.423	-3.44	0.640	193.1	0.979
0.558	0.099	-0.583	-3.28	0.541	191.8	0.965	0.479	0.156	-0.736
-2.91	0.413	190.8	0.954	0.373	0.198	-0.986	-4.35	0.337	188.6
0.930	0.202	0.278	-1.600	-1.12	0.224	183.4	0.869	0.0674	0.435
0.02	0.0926	173.0	0.728	0.0278	0.656	-0.12	0.0199	170.0	0.682
0.0134	0.704	0.78	0.0135	154.6	0.384	0.0026	0.891	1.85	0.0026
149.3	0.252	0.0017	0.938	2.10	0.0017	143.8	0.100	0.0012	0.979
2.29	0.0012	140.3	0	0.0009	1	2.50	0.0010		

Let's build a graph of the obtained dependencies:

$\eta(n_w)$

Having drawn smoothing curves, we conclude: the equation is applicable over the entire range ($n_w \in [0, 1]$).

$\eta, \text{Pa} \cdot \text{s}$



B1 Using the equation (2), determine the glass transition temperature of the solution T_g .

0.50

Let's express w_g :

$$w_g = \frac{m_g}{m_g + m_w} = \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w},$$

then:

Answer:

$$T_g = \left(140.3 + 35.272 \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} - 3.879 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^2 + 23.467 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^3 \right) K \approx 162.8 K$$

B2 Propose a linearization to determine the values of the parameters D and T_0 . After plotting the linearized dependence, determine the values of these parameters.

2.00

The dependence $\frac{1}{\ln \frac{\eta}{\eta_0}}(T)$ is linear with slope $k = \frac{1}{DT_0}$ and intercept term $b = -\frac{1}{D}$. Let's recalculate the point using

this formula and plot the linearized dependence:

$$T, ^\circ C \quad \eta, Pa \cdot s \quad T, K \quad \frac{1}{\ln \frac{\eta}{\eta_0}} \quad 1 \quad 0.0201 \quad 274 \quad 0.556 \quad 4 \quad 0.0173 \quad 277 \quad 0.607 \quad 11 \quad 0.0128 \quad 284 \quad 0.743 \quad 15 \quad 0.0104 \quad 288 \quad 0.878 \quad 35 \quad 0.0046$$

$$308 \quad 3.10 \quad 44 \quad 0.0034 \quad 317 \quad 48.1 \quad 50 \quad 0.0031 \quad 323 \quad -14.0 \quad 56 \quad 0.0026 \quad 329 \quad -4.04 \quad 60 \quad 0.0022 \quad 333 \quad -2.41 \quad 67 \quad 0.0020 \quad 340 \quad -1.96 \quad 71 \quad 0.0017$$

$$344 \quad -1.487 \quad 76 \quad 0.00145 \quad 349 \quad -1.20 \quad 85 \quad 0.0013 \quad 358 \quad -1.06 \quad 90 \quad 0.0011 \quad 363 \quad -0.903 \quad 96 \quad 0.0010 \quad 369 \quad -0.831 \quad 100 \quad 0.0009 \quad 373 \quad -0.764$$

Let's plot the resulting relationship:

$$\frac{1}{\ln \frac{\eta}{\eta_0}}(T)$$

From the proposed equation it follows that it can only be fulfilled at sufficiently low temperatures. It is clear from the graph that the equation is applicable for the first 4 points of the graph (at $T < 288 \text{ K}$). Let's demonstrate this clearly:

$$\frac{1}{\ln \frac{\eta}{\eta_0}}(T) \text{ at } T < 288 \text{ K (not required from participants)}$$

Having determined the parameters of the approximating straight line, we obtain a system of equations:

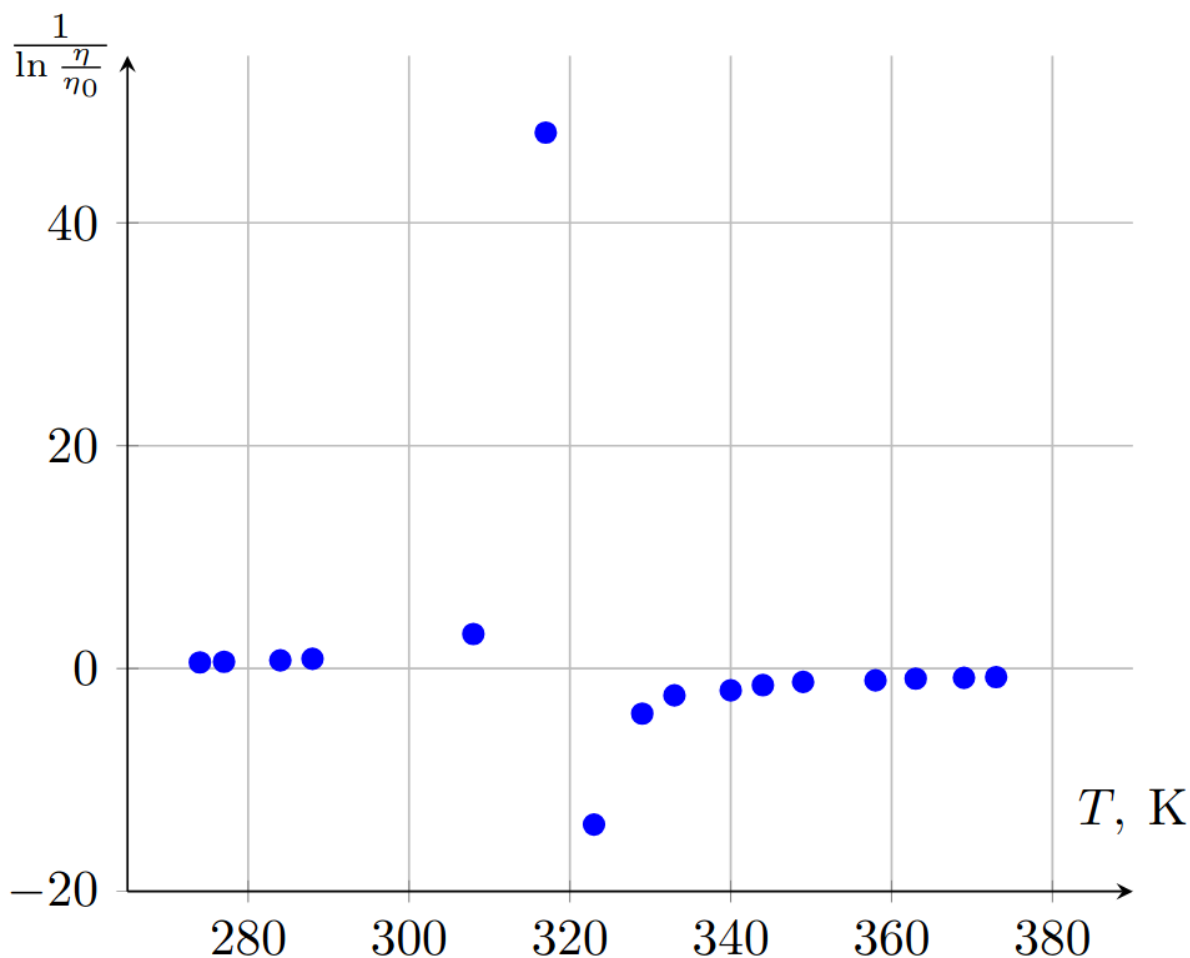
$$\begin{cases} -\frac{1}{D} = -5.60 \\ \frac{1}{DT_0} = 0.022 K^{-1}, \end{cases}$$

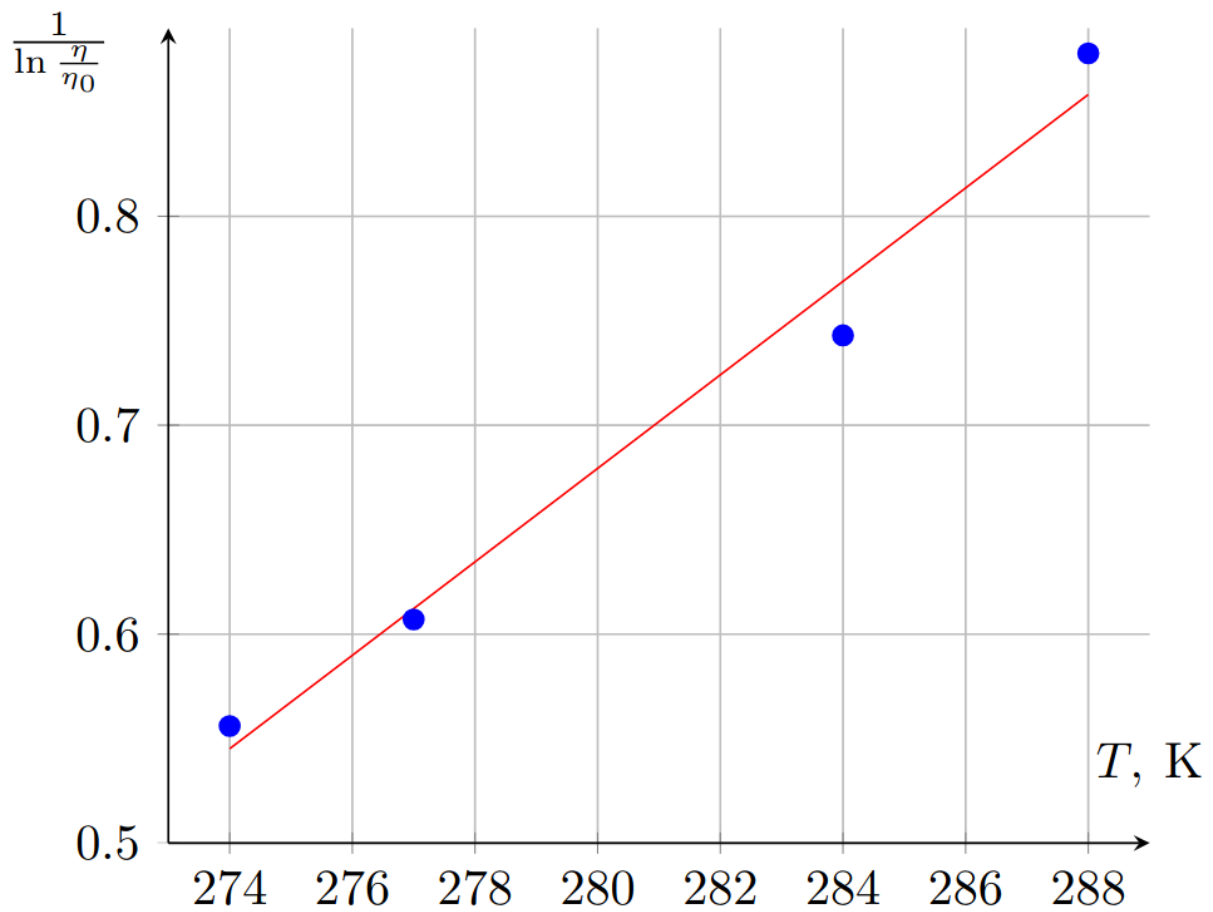
which follows:

Answer:

$$T_0 = 255 \text{ K}$$

$$D = 0.18$$





C1 Determine the viscosity of the mixture η_n at the volume concentration of glycerol $c_g = 0.5$ and temperature $T = 25^\circ\text{C}$ using the dependence on the molar concentration of water.

0.40

The equation obtained for $\eta_n(n_w)$ is applicable over the entire range. Therefore, it is enough to convert c_g into n_w :

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)} = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} (\mu_g - \mu_w)} \approx 0.80$$

and substitute it into the resulting equation. Thus we get:

Answer: $\eta_n = 6.0 \text{ mPa} \cdot \text{s}$

C2 Determine the viscosity of the mixture η_t under the same conditions using the temperature dependence. **0.40**

The equation obtained for $\eta_t(T)$ is indispensable for $T = 25^\circ\text{C}$. You can determine the viscosity at a given temperature, for example, graphically by plotting the curve $\eta_t(T)$, but the answer with good accuracy is given by a linear approximation:

$$\eta_{25} \approx \frac{\eta_{15} + \eta_{35}}{2} = 7.5 \text{ mPa} \cdot \text{s}.$$

Answer:

$$\eta_t = 7.5 \text{ mPa} \cdot \text{s}$$

C3**Compare your results. If there are discrepancies, suggest a possible reason for them.****0.20**

Answer: The meanings of η_t and η_m are divergent. One of the main possible reasons for discrepancies in results is poorly measured temperature (for example, if the room temperature changed during the experiment), because the dependence of the viscosity of the mixture on temperature is very strong.